

prises find it challenging to manage IT due to lack of alignment with businesses, broken business processes and inability to build strong business cases for modernisation. Most carry a baggage of legacy systems, ageing infrastructure and software stacks. This IT infrastructure landscape is making it cumbersome across the technology landscape. Companies are spending a colossal amount of money to streamline their business operations."

Applications of IaaS

Application areas where IaaS can be successfully deployed are test and development, website hosting, storage, backup and recovery, Web apps, high-performance computing and big data analysis.

Some operational areas where IaaS has been deployed are:

1. Pooling server and networking resources to store data and run applications in a business
2. Hosting websites on virtual servers founded on underlying physical server resources
3. Interconnecting virtual servers through a virtualised network to enhance cloud hosting capabilities or enterprise IT infrastructure

The freely available OpenStack open source framework is increasingly being used as cloud management software. It enables customers to use a largely manufacturer-neutral cloud. Other benefits include modular structure, and automation and orchestration. OpenStack is about easy self-management. Customers are provided with a graphic Web interface to manage their IT resources.

Multi-tenancy of IaaS. This feature enables using the same software and interface to configure resources and isolate customer-specific traffic and data. Multiple users who do not share or see each other's data can share the same applications while running on the same OS, using the same hardware and data storage mechanism.

In an organisation, software installation on individual employees' desktops or on a dedicated server is complicated, because all servers or installations need to be upgraded

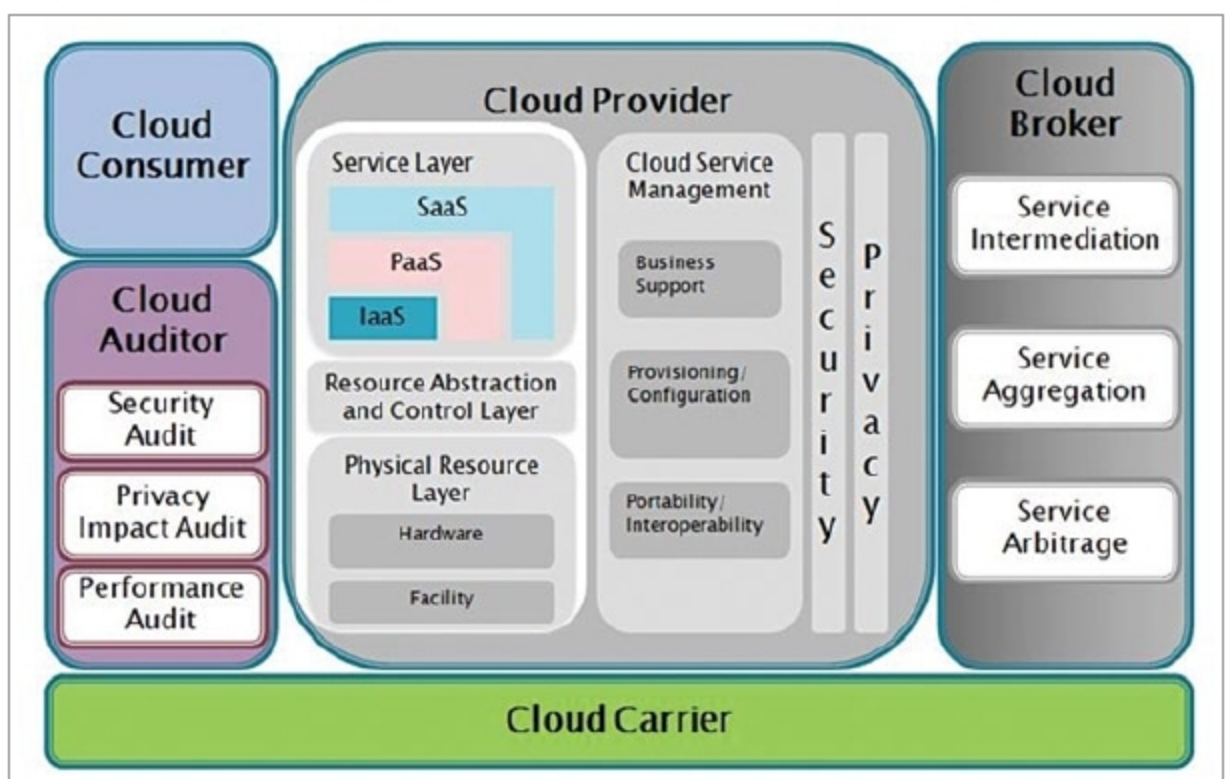


Fig. 2: Cloud computing reference architecture (Credit: IRIS Business Architect Community)

individually. Hence, servers are underused, resources are poorly utilised and maintenance becomes more complicated. Whereas, utilisation of resources improves and costs are lowered by streamlining maintenance when multiple users share same database server and applications.

Some IaaS providers are IBM, CSS Corps, Transworld Technologies, Cyient, Amazon Web Services, Cisco Metapod, Microsoft Azure and Google Compute Engine.

IaaS architecture

IaaS has a structural design that delivers computing resources via cloud as a service. IaaS architecture optimises the use of physical computing resources to maximise cost savings and revenue for an organisation. Its design depends on specific business requirements and goals.

In an IaaS model, a cloud provider hosts various infrastructure services including monitoring, detailed billing, security, log access, backup, replication, recovery, load balancing and clustering. These services are policy-driven and provide greater levels of organisation and automation to IaaS users for important infrastructure tasks, and maintaining application availability and performance.

Users are given access to virtualised components to build their own IT platforms. Computing resources provided to them include virtual server

space, bandwidth, IP addresses, network connections and load balancers. Hardware resources are received from multiple servers and networks distributed across different data centres maintained by the cloud provider.

Users can log in to IaaS platform, create virtual machines (VMs), install OSes, deploy databases, create storage for workloads and backups, and install it on that VM. They can then track costs, balance network traffic, troubleshoot application issues, monitor performance and manage disaster recovery through provider services.

Types of IaaS offerings

Bernd Hanstein, vice president - product management - IT, Rittal GmbH & Co. KG, says "Companies should factor in the requirements of data protection laws at the earliest possible stage when selecting a provider. It is important that the provider can offer a physically- (dedicated private) or logically- (virtual private) separate cloud. This is the only way to comply with strict data protection requirements and possible in-house compliance rules.

"Any company that opts to operate its IT from the cloud should select a service provider that offers and orchestrates all services from a single source. A good alternative to traditional co-located data centres and private server rooms are cloud